

NEUROSCI 366

Lecture 23

短时记忆与知觉决策 / Short-Term Memory and Perceptual Decisions

Source-aligned · Static · Searchable · Printable

Source files

- Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf

Production version: 2026-08-22

Reader profile: neuroscience undergraduate education; mathematics scaffolded from high-school algebra/trigonometry to the formal computational-neuroscience level.

This companion preserves source-page order, distinguishes source material from necessary scaffolding and extensions, and places hints/answers after the lesson. It is a static PDF: no forms, JavaScript, hidden-answer widgets, passwords, or DRM.

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1 How to use this companion

1. 先用 5-minute diagnostic 暴露前置缺口；不要立即翻答案。
2. 按原笔记页序阅读 source-page units：先看原页，再读 clean reconstruction 与补全逻辑。
3. 遇到公式按“问题 → 符号/shape/units → assumptions → 一行一步推导 → intuition → biological meaning → sanity check → failure”阅读。
4. 完成 Cumulative Knowledge Check；只在需要时依次打开 Hint 1、2、3。
5. 至少隔一个 page break 后再核对 Answer Key，并记录错误类型：concept、symbol、algebra、assumption、correlation/causation。

Source hierarchy. 原笔记是内容权威；本书中的“【必要前置】/【补全推导】/【跨课连接】/【延伸】”是为了补齐数学或迁移，不替代源材料。无法确认的内容必须进入 Uncertainty Log。

2 Learning objectives and prerequisite dependency map

- 定义并区分：cognition/decoupling。
- 从第一原则推导：DMS/Romo。
- 解释几何/计算直觉：persistent activity。
- 连接生物学解释：long recurrent mode。
- 诊断 assumptions 与 failure modes：drift-diffusion。
- 把概念迁移到新神经科学问题：psychometric/chronometric/LIP。

Dependency map

cognition/decoupling → DMS/Romo → persistent activity → long recurrent mode → drift-diffusion → psychometric/chronometric/LIP

Core question

短时记忆与知觉决策如何由 long-timescale recurrent modes、evidence accumulation 与 nonlinear attractor dynamics 实现并对应 behavior？

3 Five-minute prerequisite diagnostic

1. 本课 cognition working definition？（卡住时先看 Source-page unit 1，再看补全推导。）
2. DMS 与 parametric WM 区别？（卡住时先看 Source-page unit 2，再看补全推导。）
3. Romo task delay 中需保存什么？（卡住时先看 Source-page unit 3，再看补全推导。）
4. persistent activity 是否需要 synaptic plasticity？（卡住时先看 Source-page unit 4，再看补全推导。）
5. DDM 的 drift、noise、bound 分别影响什么？（卡住时先看 Source-page unit 5，再看补全推导。）

Scoring guide: 4–5 confident answers → proceed; 2–3 → use the prerequisite labels; 0–1 → read the formula/notation sheet once before the source-aligned lesson.

4 Source-aligned lesson / 按原笔记页序

4.1 Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf • p. 1

Introduction to cognition:

Examples of cognitive processes (non-exhaustive)

- Decision making
 - Short-term memory / working memory
 - Learning
 - Long-term memory
 - Generalization
- } Today
- } Later in unit

Cognition allows:

- The current brain to account for the past.
- The current brain to anticipate and reshape the future

Working definition: Cognition is what decouples the brain from the present.

Cognition is multifaceted and becomes more approachable when broken down.

Other topics not treated here (non-exhaustive)

- Perception
- Language
- Social behavior
- Attention

Original-note anchor. [Source: Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf, p. 1]

Clean reconstruction. Page 1: cognition working definition; decision, STM/working memory, learning/LTM/generalization.

What the note is saying. 本页的中心对象是 cognition/decoupling 与 DMS/Romo。原笔记将它们放进以下链条: cognition working definition; decision, STM/working memory, learning/LTM-generalization。

Why it follows. 准确呈现 cognition 的课程 working definition: 使脑状态不只由 present input 决定; 说明其局限。其逻辑后果是: 比较 delayed match-to-sample 与 parametric working memory; 逐步解释 Romo task 的 sample, delay, comparison, choice。后面的正式推导会逐项写出 assumptions, shape/units, limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系, 再问它表达的是定义、推导、模型预测还是经验观察; 示意图不提供未标注的定量精度。

Stop & Predict

本课 cognition working definition?

4.2 Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf • p. 2

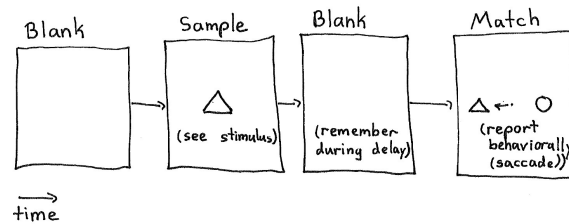
Short-term memory:

Memory: A trace of past events in the brain.

Short-term memories:

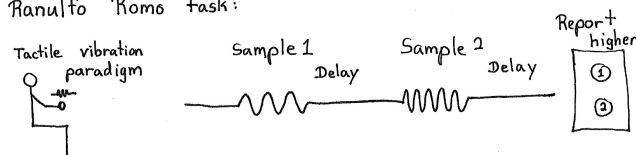
- Allows brain to account for what just happened.
- Does not require synaptic plasticity.
- Can be converted to long-term memory through synaptic plasticity.

Delayed match to sample task



Parametric working memory: store a continuous value and later use it in a computation.

Ranulfo Romo task:



Original-note anchor. [Source: Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf, p. 2]

Clean reconstruction. Page 2: delayed match-to-sample, parametric working memory, Romo vibration task.

What the note is saying. 本页的中心对象是 persistent activity 与 long recurrent mode。原笔记将它们放进以下链条: delayed match-to-sample、parametric working memory、Romo vibration task。

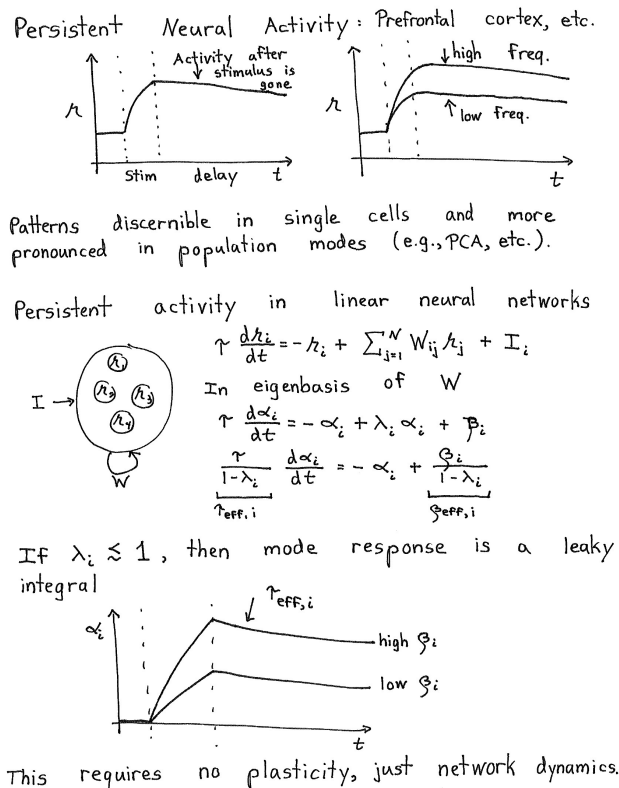
Why it follows. 比较 delayed match-to-sample 与 parametric working memory; 逐步解释 Romo task 的 sample、delay、comparison、choice。其逻辑后果是: 准确呈现 cognition 的课程 working definition: 使脑状态不只由 present input 决定; 说明其局限。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 按时间线区分 stimulus/task、recording/manipulation、measured quantity 与 inferred model variable; 结果相关性不能直接替代机制检验。

Stop & Predict

DMS 与 parametric WM 区别?

4.3 Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf • p. 3



Original-note anchor. [Source: Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf, p. 3]

Clean reconstruction. Page 3: persistent PFC activity, population modes, linear recurrent-memory dynamics.

What the note is saying. 本页的中心对象是 drift-diffusion 与 psychometric/chronometric/LIP。原笔记将它们放进以下链条: persistent PFC activity、population modes、linear recurrent-memory dynamics。

Why it follows. 解释 persistent activity 的 single-cell 与 population-mode views; 说明 STM 不必依赖当场 synaptic plasticity。其逻辑后果是: 用 recurrent eigenmode/leaky integrator 解释 graded persistent activity 与 long time constants。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系, 再问它表达的是定义、推导、模型预测还是经验观察; 示意图不提供未标注的定量精度。

Stop & Predict

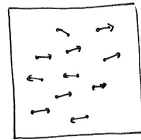
Romo task delay 中需保存什么?

4.4 Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf • p. 4

Decision Making:

The Rodo task is an example of decision making. A or B?

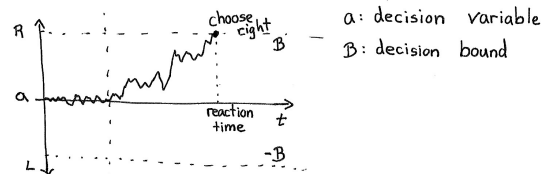
Also recall random dots task (Shadlen, Newsome)



Decision: Left or Right

Hard if coherence is low

Evidence can be accumulated over time.

Drift-Diffusion Model of behavior

Basic equation:

$$\tau_a \frac{da}{dt} = -a + \underbrace{\tau_R(t)}_{\text{R. evidence rate}} - \underbrace{\tau_L(t)}_{\text{L. evidence rate}} + \underbrace{\gamma(t)}_{\text{random noise}}$$

$$a(0) = \underbrace{b}_{\text{Bias}} \quad ; \quad a(\tau) = \pm B$$

Decision time

Many related variants: collapsing bounds? Sticky bounds?
Go cue?

Original-note anchor. [Source: Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf, p. 4]

Clean reconstruction. Page 4: random-dot decision, evidence accumulation, drift-diffusion model.

What the note is saying. 本页的中心对象是 Wong-Wang attractors 与 cognition/decoupling。原笔记将它们放进以下链条: random-dot decision、evidence accumulation、drift-diffusion model。

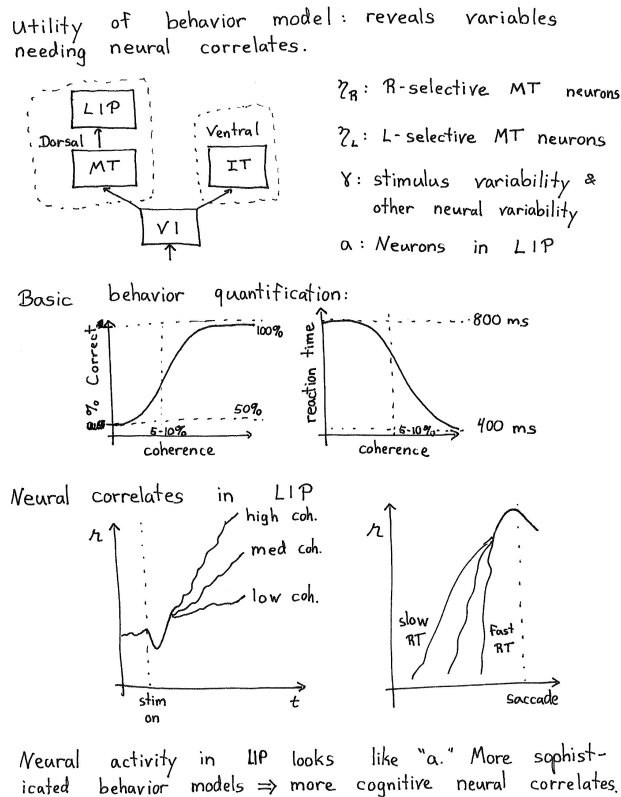
Why it follows. 逐项解释 drift-diffusion equation 的 decision variable、drift/evidence、leak、bounds、bias、noise、reaction time, 以及 collapsing/sticky bounds 等变体。其逻辑后果是: 解释 Wong-Wang reduced nonlinear recurrent model、two nullclines、stable/unstable fixed points、separatrix、weak/strong evidence 与 stochastic choice。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 按时间线区分 stimulus/task、recording/manipulation、measured quantity 与 inferred model variable; 结果相关性不能直接替代机制检验。

Stop & Predict

persistent activity 是否需要 synaptic plasticity?

4.5 Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf • p. 5



Original-note anchor. [Source: Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf, p. 5]

Clean reconstruction. Page 5: behavioral psychometric/chronometric curves, LIP neural correlates.

What the note is saying. 本页的中心对象是 DMS/Romo 与 persistent activity。原笔记将它们放进以下链条: behavioral psychometric/chronometric curves、LIP neural correlates。

Why it follows. 重构 psychometric 与 chronometric predictions; 解释 LIP ramping correlates 与 behavior-model variables 的关系, 避免 correlation=mechanism。其逻辑后果是: 逐项解释 drift-diffusion equation 的 decision variable、drift/evidence、leak、bounds、bias、noise、reaction time, 以及 collapsing/sticky bounds 等变体。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先明确横纵轴、单位、归一化与每条曲线改变的唯一参数; 区分示意趋势、模型预测和真正测得的数据点。

Stop & Predict

DDM 的 drift、noise、bound 分别影响什么?

4.6 Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf • p. 6

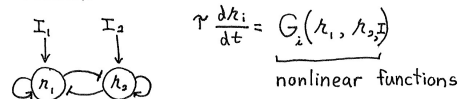
Just like short-term memory, evidence accumulation requires long integration time constants. Similar network mechanisms apply:

$$\tau_a \approx \frac{\tau}{1-\lambda}$$

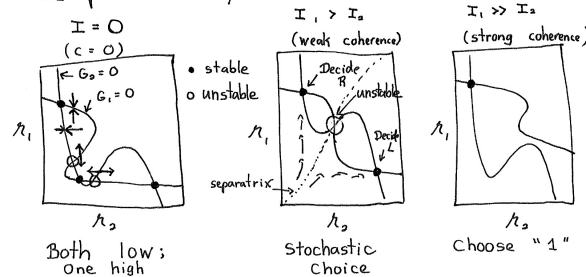
large eigenvalue

Nonlinear neural network models can also be devised for robust decision making.

Wong and Wang Model: Reduced biophysically realistic spiking neural network to 2D firing rate model.



Phase-plane analysis: Two cubic-like nullclines



Original-note anchor. [Source: Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf, p. 6]

Clean reconstruction. Page 6: long integration modes, Wong–Wang nonlinear attractor model, phase planes.

What the note is saying. 本页的中心对象是 long recurrent mode 与 drift–diffusion。原笔记将它们放进以下链条: long integration modes、Wong–Wang nonlinear attractor model、phase planes。

Why it follows. 解释 Wong–Wang reduced nonlinear recurrent model、two nullclines、stable/unstable fixed points、separatrix、weak/strong evidence 与 stochastic choice。其逻辑后果是: 用 recurrent eigenmode/leaky integrator 解释 graded persistent activity 与 long time constants。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 读图时先标出两条 nullcline/坐标轴, 再判断向量场方向、交点稳定性与轨迹穿越顺序; 轨迹本身不等于因果机制。

Stop & Predict

提高 coherence 对 RT 与 accuracy 的典型影响?

5 补全推导与数学支架 / Derivations

5.1 parametric working memory 与 leaky mode

【必要前置】 如何存储连续 sample frequency?

$$\tau \dot{\alpha} = -(1 - \lambda)\alpha + \beta(t)$$

$$\tau_{eff} = \tau / (1 - \lambda)$$

1. sample input β 把 mode amplitude 推到与 stimulus magnitude 相关的值。
2. delay 中 input ≈ 0 , state 以 τ_{eff} 衰减。
3. λ 接近 1 时可在数秒保持 graded value。
4. 不同 neurons 可能各自 mixed selectivity, 但 population mode 稳定编码。

Geometric/computational intuition. working memory content 是 slow population coordinate, 不必等于某一 neuron 持续高 firing。

Biological interpretation / failure condition. activity-silent synaptic states, dynamic coding 或 oscillatory mechanisms 也可支持 WM。

Micro-check

persistent activity 是否需要 synaptic plasticity?

5.2 leaky drift-diffusion model

【补全推导】 把 momentary left/right evidence 累积到 choice。

$$\tau_a \dot{a} = -a + [\eta_R(t) - \eta_L(t)] + \gamma(t)$$

$$a(0) = b, \quad \text{choice when } a(t) = \pm B$$

1. a 是 decision variable; $-a$ 是 leak, $\eta_R - \eta_L$ 是 signed evidence, γ 是 noise。
2. initial bias b 移动起点, asymmetric/collapsing bounds 改变 speed-accuracy tradeoff。
3. first passage to $+B/-B$ 决定 choice 与 decision time。
4. coherence 增大提高 drift, 通常 accuracy 上升、reaction time 下降。

Geometric/computational intuition. decision 是 noisy running total, 而不是读取单一瞬间 stimulus。

Biological interpretation / failure condition. urgency、sticky bounds、changing evidence 与 nondecision time 需扩展。

Micro-check

DDM 的 drift、noise、bound 分别影响什么？

5.3 psychometric/chronometric 与 neural correlates

【跨页连接】 behavioral curves 如何约束 model？

$$P(\text{right}|c) \uparrow \text{ with signed coherence } c$$

$$RT(c) \text{ typically largest near } c = 0$$

1. psychometric curve 给 choice probability vs coherence。
2. chronometric curve 给 reaction time vs coherence。
3. 同一 DDM parameters 同时预测二者, 比只拟合 accuracy 更严格。
4. LIP firing ramp 与 accumulated evidence/choice/RT 相关, 但仍可能是 readout、feedback 或 correlate。

Geometric/computational intuition. 弱 evidence 需要更长时间才随机碰到 bound, 因此慢且错误多。

Biological interpretation / failure condition. motor preparation、attention、confidence 与 task timing 可混入 recorded signals。

Micro-check

提高 coherence 对 RT 与 accuracy 的典型影响？

5.4 Wong–Wang two-attractor phase plane

【模型边界】 如何由 nonlinear recurrent competition 产生 robust discrete choices?

$$\tau \dot{r}_1 = G_1(r_1, r_2; I_1), \quad \tau \dot{r}_2 = G_2(r_1, r_2; I_2)$$

1. 两条 cubic-like nullclines 的 intersections 是 stable/unstable fixed points。
2. 无/弱 evidence 时 separatrix 附近 noise 决定进入哪个 high-rate attractor。
3. signed evidence 移动 nullclines/basins, 增加正确 attractor probability。
4. strong evidence 可消除/极度缩小错误 basin, 使 choice deterministic。

Geometric/computational intuition. integration 把 state 推离 unstable decision boundary, recurrent excitation 将 choice 固化。

Biological interpretation / failure condition. reduced model 是 biophysical network 的低维 approximation, 不是唯一决策机制。

Micro-check

提高 bound 的典型影响?

6 Cross-page synthesis / 跨页因果与数学链

本课把 cognition 暂定义为 brain state 不再只由 present input 决定。短时记忆可由 persistent population modes 保留过去 sample, 而无需当场修改 synapses。知觉决策可用 noisy evidence accumulation 与 bounds 描述 behavior; LIP ramps 是相关 neural variables, 不自动证明机制。Wong-Wang reduced attractor model 用 two-dimensional nonlinear nullclines 把 integration、competition 与 discrete choice 连接。

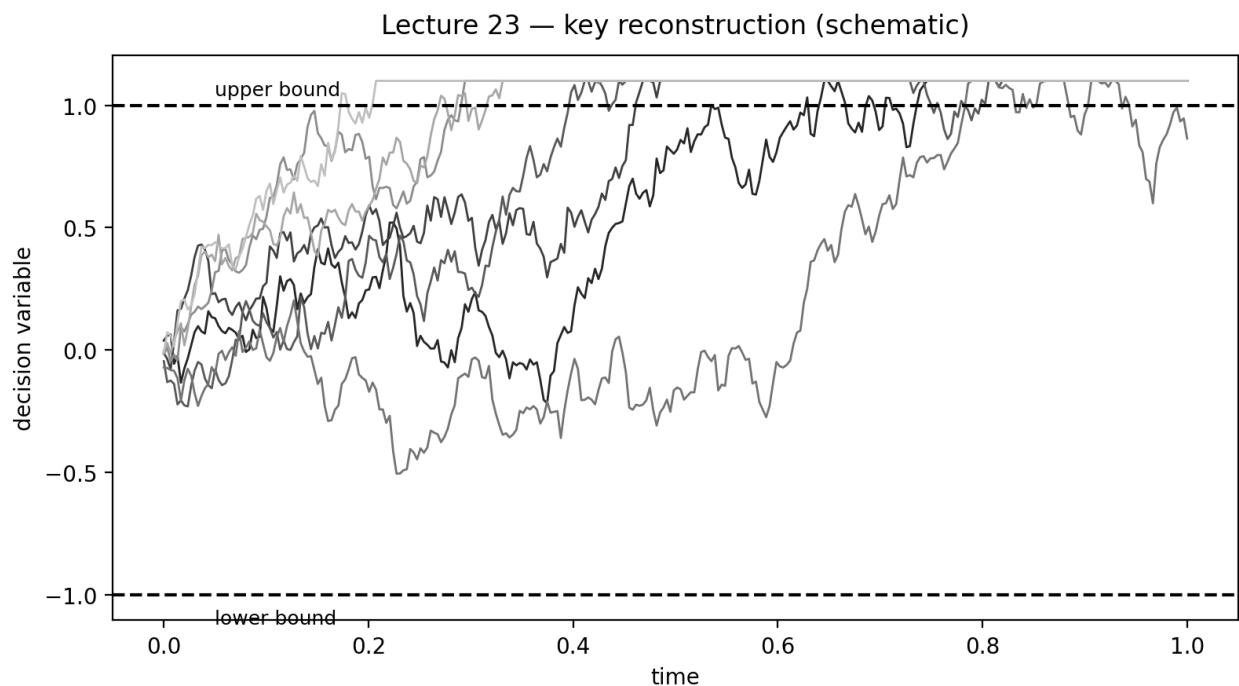


Figure note: schematic reconstruction; it encodes qualitative relations from the lesson and does not invent precise empirical data points.

7 Worked examples and transfer

7.1 Romo task sequence

【原课/核心例题】 Prompt. sample f1、delay、comparison f2、report higher。

Worked solution. delay state 必须保留 f1 的 continuous value; comparison 时计算 $\text{sign}(f2-f1)$ 并映射为 choice。

Check. 这不是简单 match/no-match; 需要 parametric memory 和 comparison computation。

7.2 DDM parameter effect

【跨课连接】 Prompt. 提高 bound B, 其他不变。

Worked solution. 通常 accuracy 提高、reaction time 变长；speed-accuracy tradeoff。
Check. 若 nondecision time 或 urgency 同时变化，观测效应可能不同。

Micro-check

Explain in your own words. 用不超过 120 字解释：本课从 source page 1 到最后一页，究竟把哪一个输入、状态、统计量或学习规则变成了可检验 prediction？

Self-reflection. 你最可能犯的是 concept、symbol、algebra、assumption 还是 causality error？写出一个具体例子。

8 Formula and notation sheet

1. persistent memory mode

$$\tau_{eff} = \tau / (1 - \lambda)$$

Conditions / conventions: $\lambda < 1$ close to 1

2. leaky evidence accumulator

$$\tau_a \dot{a} = -a + e(t) + \gamma(t)$$

Conditions / conventions: white/colored noise specified

3. decision bias

$$a(0) = b$$

Conditions / conventions: initial-condition interpretation

4. absorbing decision bounds

$$a = \pm B$$

Conditions / conventions: first-passage rule

5. psychometric function

$$P(\text{choice}|c)$$

Conditions / conventions: signed stimulus strength

6. chronometric function

$$E[RT|c]$$

Conditions / conventions: includes/nonincludes nondecision time stated

7. Wong-Wang nullclines

$$G_1 = 0, G_2 = 0$$

Conditions / conventions: 2D reduced dynamics

9 Bilingual glossary

中文	English / symbol	Operational meaning
认知	cognition	本课工作定义: brain state 部分脱离 present input, 由过去/未来目标塑造。
短时记忆	short-term memory	短时间保留刚发生 information 的能力。
工作记忆	working memory	保留并用于当前 computation 的 information。
延迟匹配样本	delayed match-to-sample	sample-delay-test 的 memory task。
持续活动	persistent activity	stimulus 结束后仍保留的 neural activity。
漂移扩散模型	drift-diffusion model, DDM	noisy evidence accumulation to bounds 的 decision model。
心理测量曲线	psychometric curve	choice/accuracy vs stimulus strength。
时间测量曲线	chronometric curve	reaction time vs stimulus strength。
分界线	separatrix	分隔不同 attractor basins 的 boundary。
吸引子	attractor	附近 states 随 dynamics 收敛的 state/set。

10 Common traps, assumptions, and limitations

1. 把 cognition 定义当作完整哲学/心理学定义。
2. 认为 STM 必须通过当场 synaptic plasticity。
3. persistent single-cell activity 与 stable population mode 混同。
4. DDM 的 decision variable 当作直接观测 neuron。
5. 只拟合 psychometric, 不检验 chronometric。
6. LIP correlation 直接等于 causal accumulator。
7. descriptive DDM 与 biophysical attractor model 混为同一层级。

Course-specific risk boundary. 区分 working-memory content、neural correlate、persistent mode 与 synaptic memory; 区分 descriptive DDM 与 biophysical attractor model。

11 Cumulative Knowledge Check

Do not turn to the Hint Bank until you have written a first attempt.

1. 本课 cognition working definition?

2. DMS 与 parametric WM 区别?

3. Romo task delay 中需保存什么?

4. persistent activity 是否需要 synaptic plasticity?

5. DDM 的 drift、noise、bound 分别影响什么?

6. 提高 coherence 对 RT 与 accuracy 的典型影响?

7. 提高 bound 的典型影响?

8. initial bias b 做什么?

9. psychometric 与 chronometric 分别画什么?

10. LIP ramp 能否证明 LIP 是唯一 accumulator?

11. Wong-Wang 中 separatrix 的作用?

12. weak evidence 下 stochastic choice 如何产生?

13. 公式表中的 “persistent memory mode” 解决什么问题? 写出适用条件与一个 sanity check。

14. 公式表中的 “leaky evidence accumulator” 解决什么问题? 写出适用条件与一个 sanity check。

15. 公式表中的 “decision bias” 解决什么问题? 写出适用条件与一个 sanity check。

16. 公式表中的 “absorbing decision bounds” 解决什么问题？写出适用条件与一个 sanity check。

17. 公式表中的 “psychometric function” 解决什么问题？写出适用条件与一个 sanity check。

18. 公式表中的 “chronometric function” 解决什么问题？写出适用条件与一个 sanity check。

19. 公式表中的 “Wong-Wang nullclines” 解决什么问题？写出适用条件与一个 sanity check。

20. 诊断并修正下面的说法：把 cognition 定义当作完整哲学/心理学定义。

12 Hint Bank

1. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
2. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
3. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
4. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
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Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
10. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
11. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
12. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
13. **Hint 1.** 先从“persistent memory mode”对应的变量关系入手。
Hint 2. 把条件“ $\lambda < 1$ close to 1”逐字转换为 assumption。
Hint 3. 写公式，再检查至少一种极端参数或单位。
14. **Hint 1.** 先从“leaky evidence accumulator”对应的变量关系入手。
Hint 2. 把条件“white/colored noise specified”逐字转换为 assumption。
Hint 3. 写公式，再检查至少一种极端参数或单位。

15. **Hint 1.** 先从 “decision bias” 对应的变量关系入手。
Hint 2. 把条件 “initial-condition interpretation” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
16. **Hint 1.** 先从 “absorbing decision bounds” 对应的变量关系入手。
Hint 2. 把条件 “first-passage rule” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
17. **Hint 1.** 先从 “psychometric function” 对应的变量关系入手。
Hint 2. 把条件 “signed stimulus strength” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
18. **Hint 1.** 先从 “chronometric function” 对应的变量关系入手。
Hint 2. 把条件 “includes/nonincludes nondecision time stated” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
19. **Hint 1.** 先从 “Wong-Wang nullclines” 对应的变量关系入手。
Hint 2. 把条件 “2D reduced dynamics” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
20. **Hint 1.** 先问: 混淆的是定义、统计估计、模型变量还是生物机制?
Hint 2. 指出一个缺失 assumption 或反例。
Hint 3. 用 “在……条件下……; 否则……” 改写。

13 Complete Answer Key with reasoning

1. 使 current brain state 不只由 present input 决定, 并受 past/future goal 影响。
2. DMS 常保留 category/item; parametric WM 保留 continuous value 并参与比较。
3. sample vibration frequency f1。
4. 不一定, 可由 recurrent dynamics/slow modes 维持。
5. drift evidence rate; noise variability; bound speed-accuracy criterion。
6. accuracy 上升, RT 下降。
7. accuracy 上升、RT 变长。
8. 把起点靠近某一 bound, 偏向对应 choice。
9. choice/accuracy vs strength; RT vs strength。
10. 不能, 可能是 correlate/readout/feedback。
11. 分隔两个 choice attractor basins。
12. state 接近 unstable boundary, noise 将其推入不同 attractor。
13. 适用条件/约定: $\lambda < 1$ close to 1。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\tau_{eff} = \tau / (1 - \lambda)$$

14. 适用条件/约定: white/colored noise specified。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\tau_a \dot{a} = -a + e(t) + \gamma(t)$$

15. 适用条件/约定: initial-condition interpretation。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$a(0) = b$$

16. 适用条件/约定: first-passage rule。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$a = \pm B$$

17. 适用条件/约定: signed stimulus strength。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$P(\text{choice}|c)$$

18. 适用条件/约定: includes/nonincludes nondecision time stated。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$E[RT|c]$$

19. 适用条件/约定: 2D reduced dynamics。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$G_1 = 0, G_2 = 0$$

20. 该说法省略了关键定义或 assumption。正确做法是先区分对象层级, 并依据本课公式/实验设计说明成立条件; 原说法不能无条件成立。

14 Source-page concordance and coverage audit

Source file	Page	Coverage status
Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf	1	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf	2	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf	3	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf	4	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf	5	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture23-Intro-ShortTermMemory-PerceptualDecisions.pdf	6	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.

15 Errata, uncertainty log, and external endnotes

15.1 Errata / source cautions

1. Page 1 的 cognition definition 明确是 working definition, 非 exhaustive。
2. Page 4 的 equation 包含 leak; 经典 perfect DDM 常无 $-a$ term, 本课采用更一般 leaky form。
3. Page 5 的 LIP ramps 是 neural correlates, 不能仅凭相似性断言机制。

15.2 Uncertainty log

No unresolved handwriting uncertainty changes a core definition, equation, figure interpretation, or answer in this companion. Where handwriting was potentially ambiguous, the source-page image is preserved and the reconstruction avoids claiming unverified detail.

15.3 External endnotes

No external web or textbook source was used to replace the uploaded notes. Added material is limited to necessary mathematical scaffolding, explicit assumptions, standard sanity checks, and clearly labeled transfer examples.

15.4 Final self-audit

All source pages listed above were visually inspected. Equations were checked for symbol definitions, units/shapes, assumptions, algebraic transitions, limiting cases, and failure conditions. The PDF is static and contains no interactive form fields or scripts.